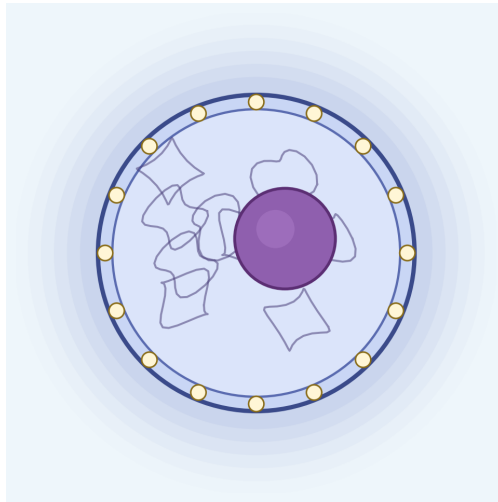


The Cell

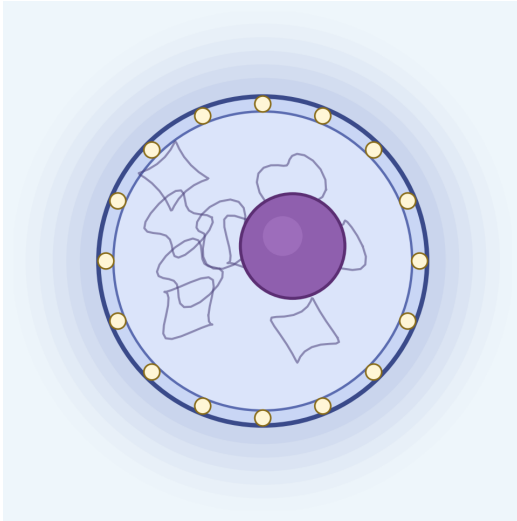
An Illustrated Guide to Important Organelles



Every living cell is a miniature, bustling city — with power plants, factories, shipping departments, recycling centers, and a command headquarters. This guide walks through the essential organelles found in eukaryotic cells, pairing a detailed illustration with a clear explanation of structure and function for each.

Nucleus

The control center of the cell



Structure & Function

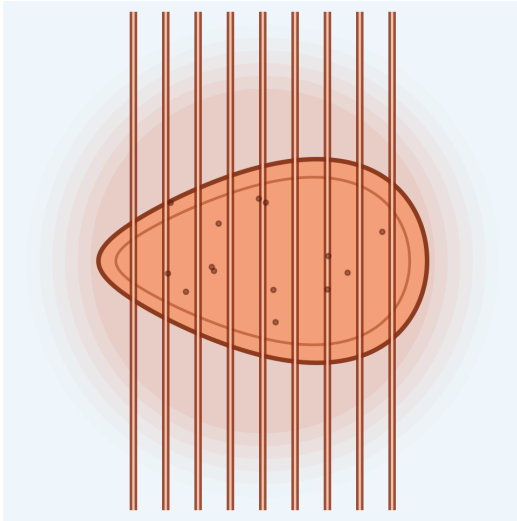
- Surrounded by a **double membrane** called the nuclear envelope, studded with nuclear pores.
- Nuclear pores regulate the movement of molecules — such as RNA and proteins — into and out of the nucleus.
- Contains **chromatin** (DNA loosely wound around proteins), which condenses into chromosomes before cell division.
- Houses the **nucleolus**, a dense region where ribosomal RNA is synthesized and ribosome subunits are assembled.
- Directs cellular activity by controlling gene expression — essentially the instruction manual for the whole cell.

Quick Facts

- Largest organelle in most animal cells.
- Discovered by Robert Brown in 1831.
- Human cells contain about 2 meters of DNA packed inside a nucleus only ~6 micrometers wide.

Mitochondrion

The powerhouse of the cell



Structure & Function

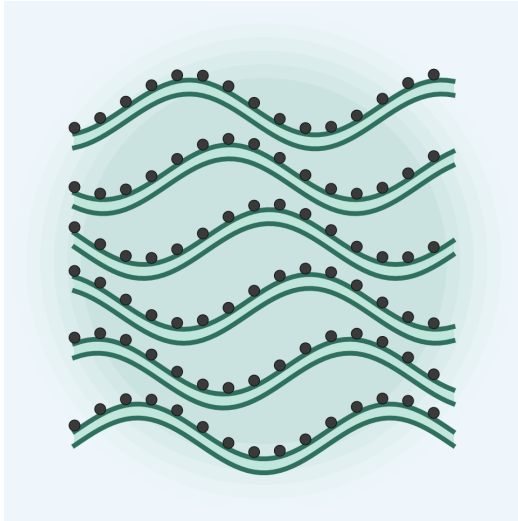
- Enclosed by a **double membrane**: a smooth outer membrane and a highly folded inner membrane.
- The folds of the inner membrane, called **cristae**, increase surface area for energy production.
- Site of **cellular respiration**, converting glucose and oxygen into usable chemical energy (ATP).
- Contains its own small circular DNA and ribosomes, a legacy of its ancient bacterial origins.
- The fluid-filled interior, or **matrix**, holds enzymes needed for the citric acid (Krebs) cycle.

Quick Facts

- Thought to have originated from ancient free-living bacteria (endosymbiotic theory).
- Number per cell varies enormously — from a few hundred to several thousand, depending on energy needs.
- Muscle and liver cells, which need lots of energy, are especially rich in mitochondria.

Rough Endoplasmic Reticulum

The protein assembly line



Structure & Function

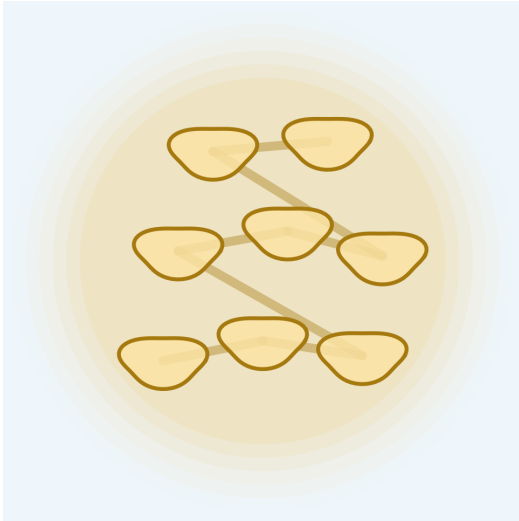
- A network of flattened membrane sacs (cisternae) continuous with the nuclear envelope.
- Studded on its outer surface with **ribosomes**, giving it a "rough" appearance.
- Ribosomes attached here synthesize proteins destined for secretion, the cell membrane, or other organelles.
- Newly made proteins are folded and can be tagged with carbohydrate chains inside the ER lumen.
- Proteins are packaged into transport vesicles and shipped onward to the Golgi apparatus.

Quick Facts

- Especially abundant in cells that secrete large amounts of protein, such as pancreatic cells.
- "Rough" refers only to appearance under a microscope — the ribosomes create the studded texture.

Smooth Endoplasmic Reticulum

The lipid and detox workshop



Structure & Function

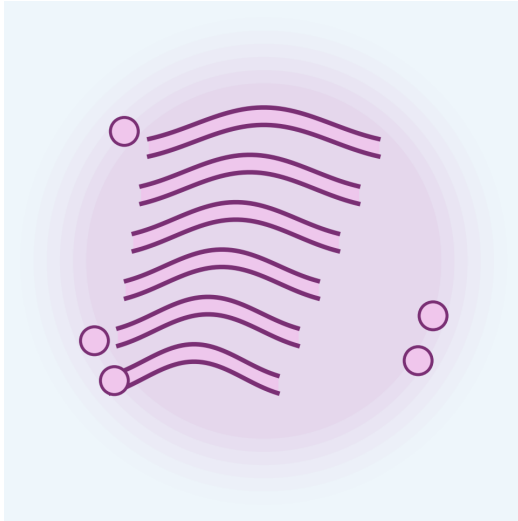
- A tubular, interconnected network of membranes lacking ribosomes — hence "smooth."
- Main site of **lipid and steroid hormone synthesis**, including cholesterol.
- Plays a key role in **detoxifying** drugs and harmful metabolic byproducts, especially in liver cells.
- Stores and regulates **calcium ions**, which is especially important in muscle cells (as the sarcoplasmic reticulum).
- Helps package and transport lipids to other parts of the cell.

Quick Facts

- Liver cells contain unusually large amounts of smooth ER due to their detox role.
- In muscle cells, a specialized form controls contraction by releasing calcium.

Golgi Apparatus

The post office of the cell



Structure & Function

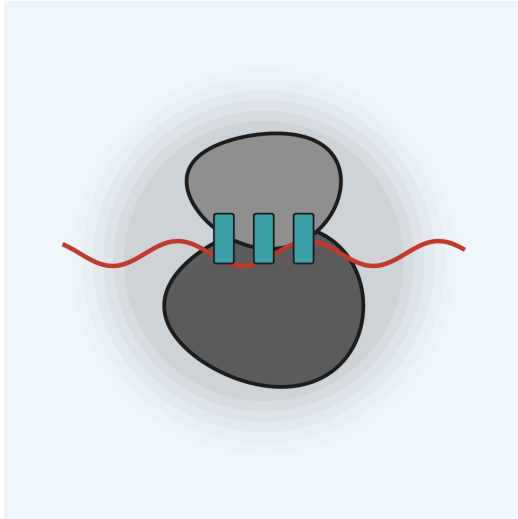
- A stack of flattened, membrane-bound sacs called **cisternae**.
- Receives proteins and lipids from the ER at its "cis" face and modifies them as they move through.
- Adds finishing touches such as sugar groups (glycosylation) and tags proteins for their final destination.
- Packages finished products into vesicles that bud off from the "trans" face for delivery.
- Also produces lysosomes in animal cells.

Quick Facts

- Named after Camillo Golgi, who first described it in 1897.
- A single cell may contain anywhere from one to hundreds of Golgi stacks.

Ribosome

The protein factory



Structure & Function

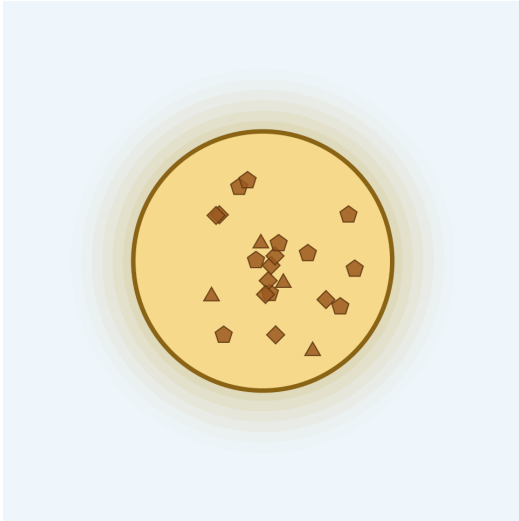
- Composed of two subunits — one large, one small — made of ribosomal RNA (rRNA) and proteins.
- Reads the genetic code carried by **messenger RNA (mRNA)** and translates it into a chain of amino acids.
- Transfer RNA (tRNA) molecules deliver the correct amino acids in the order specified by the mRNA sequence.
- Found **free-floating** in the cytoplasm or attached to the rough endoplasmic reticulum.
- Not membrane-bound — technically not a true organelle by some definitions, but essential all the same.

Quick Facts

- A single cell can contain millions of ribosomes.
- Found in every domain of life — bacteria, archaea, and eukaryotes — a sign of their ancient origin.

Lysosome

The recycling and waste center



Structure & Function

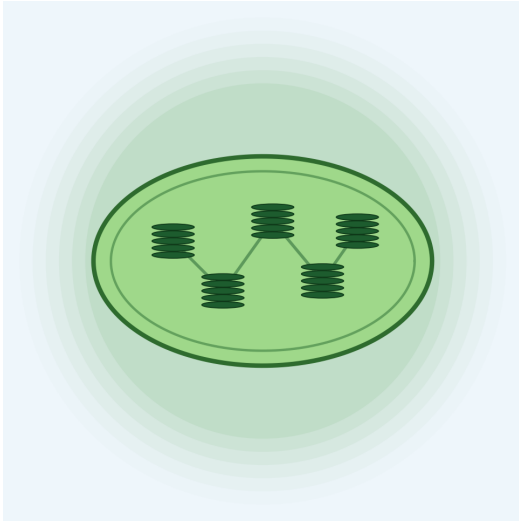
- A membrane-bound sac filled with roughly 40-50 different **digestive enzymes**.
- Breaks down worn-out organelles, food particles, and foreign invaders like bacteria.
- Maintains an acidic internal environment (around pH 4.5) that activates its enzymes.
- Plays a role in **programmed cell death** (apoptosis) by releasing enzymes when triggered.
- Recycles cellular building blocks — sugars, amino acids, and nucleotides — back into the cytoplasm for reuse.

Quick Facts

- Sometimes called "suicide bags" because of their role in breaking down cell components.
- Several genetic disorders (lysosomal storage diseases) result from missing or faulty lysosomal enzymes.

Chloroplast

The solar power plant of plant cells



Structure & Function

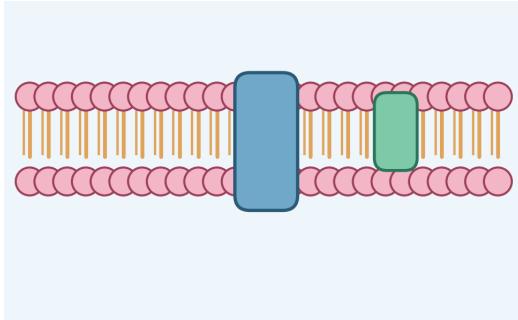
- Found in plant and algal cells; enclosed by a **double membrane**.
- Contains stacks of disc-shaped membranes called **thylakoids**, grouped into stacks called grana.
- Thylakoid membranes hold **chlorophyll**, the pigment that captures light energy and gives plants their green color.
- Site of **photosynthesis** — converting light energy, water, and carbon dioxide into glucose and oxygen.
- The fluid surrounding the thylakoids, the **stroma**, is where sugar-building reactions (the Calvin cycle) occur.

Quick Facts

- Like mitochondria, chloroplasts have their own DNA and likely arose from ancient photosynthetic bacteria.
- A single leaf cell can contain 30–50 chloroplasts.

Plasma (Cell) Membrane

The gatekeeper of the cell



Structure & Function

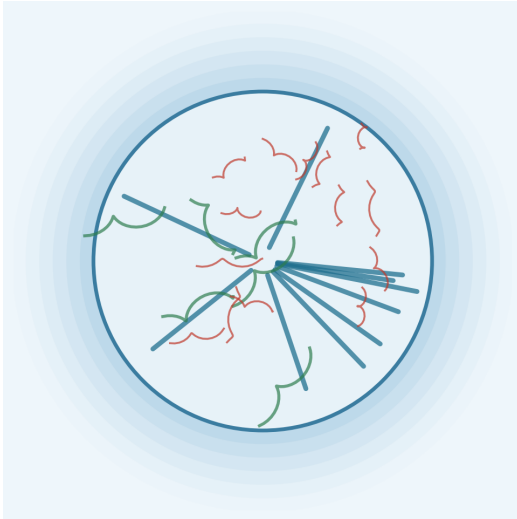
- Built from a **phospholipid bilayer** — two layers of fat molecules with water-loving heads and water-fearing tails.
- Selectively permeable: controls what enters and exits the cell.
- Embedded **membrane proteins** act as channels, pumps, receptors, and identification markers.
- Cholesterol molecules woven throughout help maintain membrane flexibility and stability.
- Described by the **fluid mosaic model** — components can drift and move within the layer.

Quick Facts

- Roughly 5–10 nanometers thick — thousands of times thinner than a human hair.
- Present in every living cell, from bacteria to humans.

Cytoskeleton

The scaffolding and transport network



Structure & Function

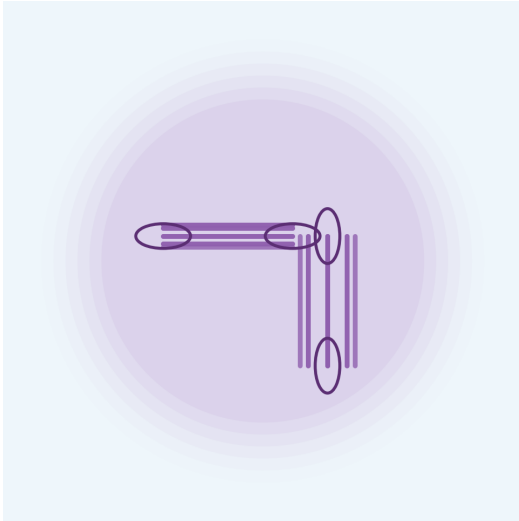
- A dynamic internal framework of protein filaments giving the cell shape, strength, and organization.
- **Microtubules** (thick, blue) act as tracks for organelle transport and form the mitotic spindle during division.
- **Microfilaments** / actin filaments (thin, red) support cell shape and enable movement and muscle contraction.
- **Intermediate filaments** (green) provide mechanical strength and anchor the nucleus and other organelles in place.
- Constantly assembles and disassembles to allow the cell to change shape, divide, and move.

Quick Facts

- Microtubules are built from a protein called tubulin; microfilaments from actin.
- Motor proteins like kinesin and myosin "walk" along these filaments to transport cargo.

Centrioles

The organizers of cell division



Structure & Function

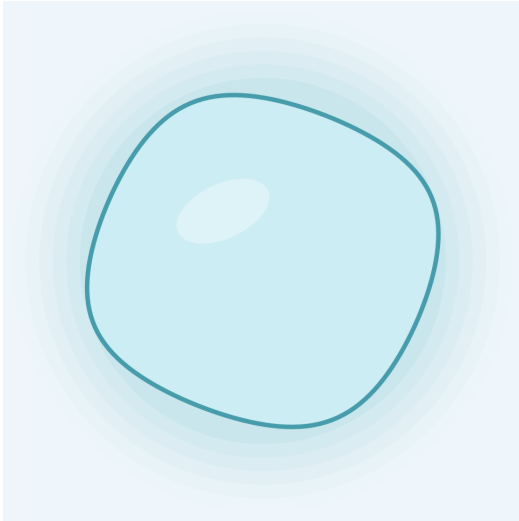
- Two cylindrical structures, arranged at right angles to each other, forming the **centrosome**.
- Each centriole is built from nine triplets of microtubules arranged in a ring.
- Organize the **mitotic spindle**, which pulls chromosomes apart during cell division.
- Also give rise to the basal bodies that anchor cilia and flagella.
- Duplicate once per cell cycle so each daughter cell inherits its own pair.

Quick Facts

- Absent in most plant cells and many fungi, which organize spindles differently.
- First described in the late 1800s using early light microscopy.

Vacuole

The storage tank of the cell



Structure & Function

- A large, fluid-filled sac enclosed by a single membrane called the **tonoplast**.
- In plant cells, a **central vacuole** can occupy up to 90% of the cell's volume.
- Stores water, nutrients, ions, and waste products; helps maintain internal pressure (turgor) that keeps plants rigid.
- In animal cells, smaller vacuoles assist with storage, waste removal, and osmotic regulation.
- In some protists, contractile vacuoles actively pump out excess water to prevent bursting.

Quick Facts

- Turgor pressure from the central vacuole is what keeps a healthy plant standing upright.
- When a plant wilts, its vacuoles have lost much of their water.